in the material, govern the appearance of the observed spectra. Absorption of radiation in this regime is often accompanied by electronic relaxation processes, and, depending on the nature of the electronic states involved, can cause fluorescence and/or phosphorescence, providing another window on the chemical makeup and structure of the material.

Visible and ultraviolet spectroscopies are frequently coupled with other measurement techniques to provide additional insight into the makeup and nature of materials. Changes in materials induced by electrochemical reactions, for instance, can change the observed spectra, or even lead directly to chemiluminescence (ELECTROCHEMICAL TECHNIQUES). Of particular interest to semiconductor studies is the luminescence (photoluminescence, or PL) that follows the promotion of electrons into the valence band of the material by incident photons (ELECTRICAL AND ELECTRONIC MEASUREMENTS). Electrons may even be ejected completely from the material, providing a measure of the work function (the photoelectric effect).

At the high end of the ultraviolet range (vacuum UV) and beyond, yet another shift in convention occurs. This regime traditionally employs the unit of electron volts (eV). Phenomenology of interactions ranges from inner-shell electronic transitions, as observed in ultraviolet and x-ray photoelectron spectroscopy, to nuclear rearrangement, as observed in gamma ray emission in nuclear reactions (ION-BEAM TECHNIQUES). Emitted electrons themselves carry spectroscopic information as kinetic energy; this is the basis for ESCA (electron spectroscopy for chemical analysis), energy dispersive spectroscopy, and Auger electron spectroscopy (ELECTRON TECHNIQUES).

The Mössbauer effect relies on the recoil-free emission of nuclear transition gamma radiation from the nuclei in a solid (the source) and its resonance absorption by identical nuclei in another solid (the absorber). Because recoilless resonance absorption yields an extremely sharp resonance, Mössbauer spectroscopy is a very sensitive probe of the slightest energy level shifts and therefore of the local environment of the emitting and absorbing atoms.

The units collected in this chapter present methods that are useful in the optical characterization of materials. The preceding discussion illustrates how spectroscopy permeates the entire *Characterization of Materials* volume—called upon where appropriate to specific materials studies. Spectroscopy, as a general category of techniques, will certainly remain central to the characterization of materials as materials science continues its advance.

ALAN C. SAMUELS

OPTICAL MICROSCOPY

INTRODUCTION

Among the numerous investigative techniques used to study materials, optical microscopy, with its several diverse variations, is important to the researcher and/or materials engineer for obtaining information concerning the structural state of a material. In the field of metallurgy, light optical metallography is the most widely used investigative tool, and to a lesser degree its methods may be extended to the investigation of nonmetallic materials. The state of microstructure of a metal or alloy, or other engineering material, whether ceramic, polymer or composite, is related directly to its physical, chemical, and mechanical properties as they are influenced by processing and/or service environment. To obtain qualitative, and in many cases quantitative information about the microstructural state of a material, it is often necessary to perform a sequence of specimen-preparation operations, which will reveal the microstructure for observation. Selecting a specimen, cutting it from the bulk material, grinding and polishing an artifact-free surface, then etching the surface to reveal the microstructure, is the usual sequence of specimen-preparation operations (see SAMPLE PREPARATION FOR METALLOGRAPHY). This scheme is appropriate for metals and alloys as well as some ceramic, polymer, and composite materials. Specimens thus prepared are then viewed with a reflected-light microscope (see REFLECTED-LIGHT OPTICAL MICROSCOPY). The microstructure of transparent or translucent ceramic and polymer specimens may be observed using a transmitted-light microscope when appropriately thinned sections are prepared. In all cases, observation of the microstructure is limited by the relatively shallow depth of focus of optical microscopes, the depth of focus being inversely related to the magnification.

Optical microscopy as related to the study of engineering materials can be divided into the categories of reflected-light microscopy for opaque specimens and transmitted-light microscopy for transparent specimens. On the whole, reflected light microscopy is of more utility to materials scientists and engineers because the vast bulk of engineering materials are opaque. Within each of the categories, there are a number of illumination variations (bright field, dark field, polarized light, sensitive tint, phase contrast, and differential interference contrast), each providing a different or enhanced observation of microstructural features. It is fortunate that a majority of illumination variations of reflected-light microscopy parallel those of transmitted-light microscopy.

In general, one does not set out to employ optical microscopy to measure a material property or set of properties. Information gained using optical microscopy is more often than not of a qualitative nature, which, to a skilled observer, translates to a set of correspondences among a material's physical, mechanical, chemical properties as influenced by chemical composition, processing and service history. One can use the measurement methods of quantitative microscopy to follow the progress of microstructural evolution during processing or during the course of exposure to a service environment. Such techniques may be used to follow grain growth, the coarsening of precipitates during a process, or the evolution of microstructure that would lead to catastrophic failure under service conditions. Such a case might be the evolution of the microstructure of steam pipes carrying superheated steam at high pressure that will ultimately fail by creep rupture.

In comparison with other instruments used to produce images of microstructure—transmission electron microscope (TRANSMISSION ELECTRON MICROSCOPY and SCANNING TRANSMISSION ELECTRON MICROSCOPY), scanning electron microscope (SCANNING ELECTRON MICROSCOPY), field ion microscope, scanning tunneling microscope (SCANNING TUNNELING MICROSCOPY), atomic force microscope, and electron beam probe—the light-optical microscope is the instrument of choice for a majority of studies. The reasons are as follows: (1) instrument cost is relatively low, (2) less technician training is required, (3) specimen preparation is relatively straightforward, and (4) the amount of information that can be obtained by an experienced microscopist is large. Because the maximum resolution is limited by the wavelength of light, useful magnification is limited to ~1600 times. At these upper limits of resolution and magnification, microstructural features as small as ~325 nm may be observed. Most of the other instruments are better suited to much higher magnifications, but the area of specimen surface is severely limited, so sampling of the microstructure state is better using optical microscopy.

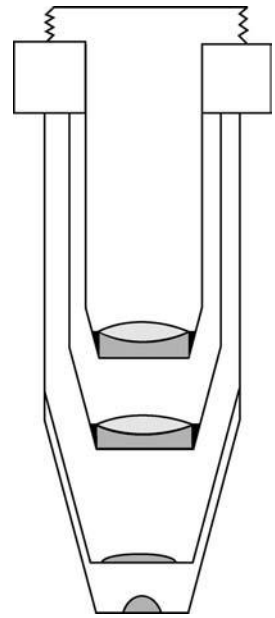


Figure 1. Schematic diagram of a typical objective lens. Note that the plano-convex lens is the "front" lens element of the objective and faces the specimen.

PRACTICAL ASPECTS OF THE METHOD

The Basic Optical Microscope

Early microscopes, though rudimentary compared to today's instruments, provided suitable magnifications, image resolution, and contrast to give early researchers and engineers suitably detailed visual information necessary for the advancement of science and materials application. Modern instruments are indeed sophisticated and incorporate a number of devices designed to enhance image quality, to provide special imaging techniques, and to record images photographically or digitally. Regardless of the sophistication of an instrument, the fundamental components that are of particular importance are a light source, an objective lens, and an ocular or eyepiece.

Significant development of the microscope was not forthcoming until the wave nature of light became known and Huygens proposed a principle that accounted for the propagation of light based on an advancing front of wavelets (Jenkins and White, 1957). Rayleigh was the first to develop a criterion for resolution, which was later refined by Abbe, who is often credited as the "father of optical microscopes" (Gifkins, 1970). Abbe's theory of the microscope was founded upon the concept that every subject consisted of a collection of diffracting points like pinholes, each pinhole behaving like a Huygens light source (Jenkins and White, 1957). It is the function of the objective lens to collect diffracted rays from an illuminated specimen to form an image by constructive and destructive interference from a collection of Huygens sources. Therefore, the function of a lens through which the specimen is viewed is to gather more diffraction information than could be gathered by the unaided eye. This means that an objective lens must be of such geometric configuration as to gather as many orders of diffracted light as possible, to give an image of suitable resolution. The more orders of diffracted rays that can be collected by the objective lens, the greater is the resolution of the lens, and generally the higher the magnification attainable.

By its most basic definition, a lens is a piece of glass or other transparent material bounded by two surfaces, each of a different and/or opposite curvature. From a geometric point of view, light rays passing through a lens either converge or diverge. Any device that can be used to concentrate or disperse light by means of refraction can be called a lens. Objective lenses usually contain a first element (nearest the specimen) that is plano-convex, having a plane front surface and a spherical rear surface (Fig. 1). Additionally, the objective lens may contain a number of elements arranged in several groups, secured in a lens barrel to keep the elements and groups in proper spatial relationship to one another. Likewise the eyepiece or ocular (Fig. 2), usually consists of two lenses, or two or more groups of lenses, each group containing two or more elements.

In a microscope the objective lens, acting like a projector lens, forms a real image of the subject near the top of the microscope tube. Because the image is formed in air, it is often called the aerial image (Restivo, 1992). The eyepiece acts to magnify this real image as a virtual image which is projected by the cornea on the retina of the eye, or it may be projected on the plane of a photographic film or on the surface of some other image recording device. The final magnification observed is the product of the magnification of the objective lens times the magnification of the ocular, except when the microscope contains a zoom magnification device or a magnification multiplier.

To a microscopist, the properties of the objective and the ocular lenses are a first consideration in setting out to use a microscope; therefore, two important properties of objective lenses, magnification and numerical aperture, are usually engraved on the barrel of objective lenses. Like-

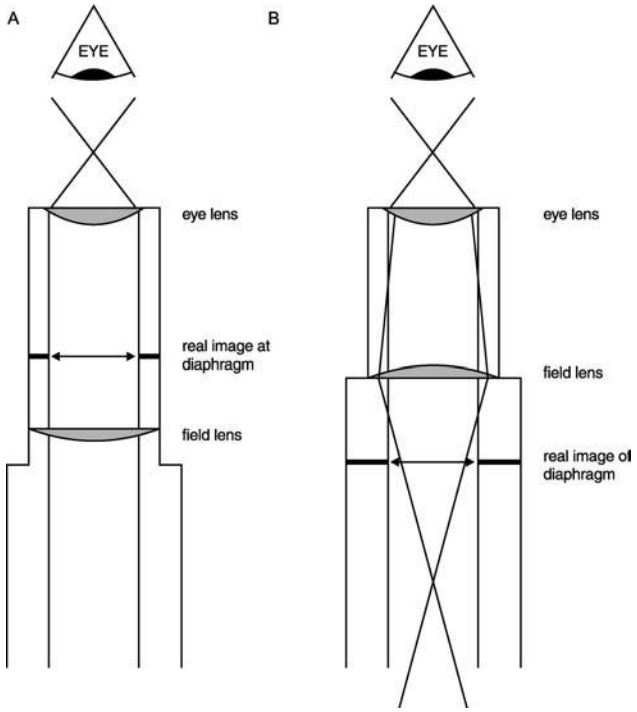


Figure 2. Schematic diagrams of (**A**) a Huygenian ocular and (**B**) a Ramsden ocular. Note the relative orientation of the field lens and the field diaphragms.

wise, magnification is engraved on the ocular. For an objective and ocular to work together properly, their focal lengths must be such that they are separated by a specific distance, so that a virtual image is produced at the eye or a real image is produced at the viewing plane. This distance is the tube length, defined as the distance between the rear focal plane of the objective and the front focal plane of the ocular. It is universally accepted that for any given microscope the objectives and oculars are designed to work together. Objectives and oculars designed for an instrument must be designed for the same tube length. As a practical rule, objective lenses and oculars should not be mixed or transferred to other microscopes. A number of recent instruments are infinity-corrected, meaning that the eyepieces must also be infinity corrected. Infinity-corrected systems allow for variations in the length of the optical path, so that other devices can be inserted for special purposes.

Characteristics of the Microscope

The characteristics of the microscope that are of the greatest importance to the user are the magnification, resolution, and the absence of lens aberrations.

While *magnification* appears to be a simple concept, without concurrent consideration of resolution it becomes relatively unimportant. Magnification alone cannot assist in revealing detail unless the system also possesses resolving power. The microscopist defines resolution as the distance between two closely spaced points in the subject that can be made observable; thus, resolution, expressed as a linear measure, is better the smaller the distance between the points. Attainable resolution is established by the properties of the objective lens, the wavelength of the illumination, and the index of refraction of the medium between the objective lens and the specimen. When the magnification is great enough for the resolution to be perceived by the eye, no amount of additional magnification will cause the image to have more resolution; no more detail will be visible. Regardless of the resolution of photographic films or charge-coupled devices, the resolution of the microscope, primarily that of the objective lens, is controlling in terms of image detail that can be detected.

Calibration of magnification is altogether straightforward. A ruled scale, usually in tenths and hundredths of a millimeter, is placed on the stage of the microscope and brought into sharp focus. It is customary to photograph such a scale at the full series of magnifications of which the instrument is capable. By measuring the photographic images of the magnified scale, one can easily determine the precise magnification.

The degree of resolution is dependent upon the amount of diffraction information that can be gathered by the objective lens. This factor is most conveniently expressed in terms of the numerical aperture (N.A.) of the objective lens. The N.A. of an objective lens is given by $\text{N.A.} = n \sin \mu$, where n is the index of refraction of the medium between the specimen and the lens (for air, $n = 1$), and μ is the half-angle of the cone of light capable of being accepted by the objective (Gifkins, 1970).

Resolution (R) is dependent upon the wavelength of light (λ) illuminating the specimen, and the N.A. of the objective lens: $R = \lambda/(2 \times \text{N.A.})$ (Gifkins, 1970). The larger the numerical aperture and/or the shorter the wavelength of light, the finer the detail that can be resolved. This expression for resolution is used expressly for reflected-light microscopy, where the objective lens carries light to the specimen, then acts again to collect light reflected from the specimen. Resolution of other reflected-light microscopes, particularly those that operate at low magnification and utilize an off-axis external light source, is not a major concern, because the low-power objectives have small N.A. values.

In transmitted-light microscopy a separate condenser lens supplies light through the specimen; thus, the expression becomes $R = \lambda/(\text{N.A.}_{\text{condenser}} + \text{N.A.}_{\text{objective}})$. By way of example, if one illuminates a specimen with green light ($\lambda = 530$ nm) and uses an objective with a 0.65 N.A., then the $R = 407.7$ nm (4.077×10^{-4} mm). R is the distance between two points on the specimen that can be resolved by the microscope. Even though these points may be resolved by the microscope, they will not appear as separate points unless the magnification of the microscope is sufficient to place them within the resolving power of the human eye. Most people can detect an angular separation of from 1 to 2 min of arc. This amounts to a distance between points of $\sim$0.11 mm at a viewing distance of 250 mm. Therefore, by dividing 0.11 mm by 4.077×10^{-4} mm, one obtains a system magnification of $\sim 270\times$. At least this amount of magnification is required in order to make the two points perceptible to the eye. Any magnification greater than this only makes the image more easily perceived by the eye, but the resolution is not further enhanced.

In practice, the largest numerical aperture is ~1.6. Using this value for numerical aperture and again using green light, one finds that the maximum resolution occurs at a magnification of the order of 650×. A magnification of 1000 or 1600 times may be justified by the increased ease of perception. Some microscope manufacturers suggest, by "rule of thumb," that the practical magnification should be 1000 times the numerical aperture. Other sources recommend lower magnifications depending on the type and magnification of the objective.

Because the resolving power of a microscope is improved as the wavelength of the illumination decreases, it is evident that blue light will give slightly better resolution than green light. Wavelengths of <400 nm have been used successfully to enhance resolution. Because conventional optical glasses absorb wavelengths below ~300 nm, the use of special optical elements of quartz, fluorite, calcite, or combinations thereof are required. Thus, all wavelengths down to ~200 nm can be transmitted through the microscope. Mercury vapor lamps produce a large amount of short-wavelength illumination and have proven suitable for ultraviolet (UV) microscopy. Because of the shortness of the wavelength, images can be damaging to the eye if viewed directly; therefore the UV part of the spectrum must be filtered out. Photomicroscopy presents a significant problem in focusing the image. If the image is focused at the film plane with the ultraviolet wavelengths filtered out, the image produced by the shorter wavelengths will not be in focus at the film plane. In 1926, Lucas developed a method for focusing the image on a fluorescent screen. Although resolution can theoretically be enhanced by as much as 60%, a more practical enhancement of 33% can be obtained using a wavelength of 365 nm rather than 540 nm (Kehl, 1949). Ultraviolet microscopy has not become a major technique in materials science because of the great amount of skill required to produce exceptionally flat and artifact-free surfaces. Also, the development of scanning electron micro-scopy (see SCANNING ELECTRON MICROSCOPY) has made it possible to get exceptional resolution, with great depth of field, without great concern for sample preparation.

A general knowledge of lens aberrations is important to the present-day microscopist. While some modern instruments are of very high quality, the kinds of objectives and oculars that are available are numerous, and it must be understood what aberrations are characteristic of each. In addition, various oculars (or projector lenses) are available for correction of aberrations introduced by objective lenses.

Lens aberrations are of two general classes: spherical and chromatic (Gifkins, 1970; Vander Voort, 1984; Kehl, 1949). Spherical aberrations are present even with monochromatic illumination. Its subclasses are spherical aberration itself, coma, curvature of field, astigmatism, and distortion. Chromatic aberration is primarily associated with objectives.

Spherical aberration occurs when a single lens is ground with truly spherical surfaces. Parallel rays come to focus at slightly different points according to their path through the lens—through the center portion of the lens, or through its outer portions. Thus, the image of a point source at the center of the field becomes split into a series of overlapping point source images resulting in an unsharp circle of confusion instead of a single sharp point source image. For spherical positive (convex or convergent) lenses, the spherical aberration is exactly the opposite of that produced by a spherical negative (concave or divergent) lens ground to the same radius of curvature. Correction for spherical aberration can take two forms. The first involves the use of a doublet (two lenses cemented together—one of positive curvature and one of negative curvature). The development of lens formulas for a doublet is rather more complex than it would seem, for should both the positive and negative lenses of the doublet be ground to the exact same radius, then the magnification of the doublet would be 1×. A second method of correcting for spherical aberration is to use a single lens with each face ground to a different radius; such a lens is said to be aspherical.

Coma affects those portions of the image away from the center. Differences in magnification resulting from ray paths that meet the lens at widely differing angles cause the image of a point to appear more like a comma. Lens formulas for the correction of coma provide for the sine of the angle made by each incident ray with the front surface of a lens to have a constant ratio to the sine of each corresponding refracted ray.

Curvature of field arises because the sharpest image of an extended object lies on a curved surface rather than a flat plane. While slight curvature of field is not a serious problem for visual work, because the eye can partially adjust for the visual image and fine focus adjustments can overcome that for which the eye cannot compensate—it becomes a serious problem for photographic work. At high magnifications, the center of the image may be sharp, with sharpness falling off radially away from the center. Attempts to focus the outer portions of the image only cause the central portion of the image to go out of focus. Correction for curvature of field is done by introducing correction in the ocular with the addition of specially designed lens groups. For this reason, it must be understood that oculars are designed to work with a specific objective or set of objectives. One should not attempt to use oculars in another instrument or with objectives for which they are not designed.

Astigmatism is a defect that causes points to be imaged as lines (or discs). Lenses designed to correct for astigmatism are called anastigmats. Correction for astigmatism is done by introducing an additional departure from spherical lens surfaces, together with the matching of refractive indices of the lens components.

Distortion is mainly an aberration found in oculars, where straight lines appear as curved lines in the outer portion of the image. The curvature may be either positive or negative and can often be observed when viewing the image of a stage micrometer. Again it is produced by unequal magnifications from various zones of the lenses in an ocular. Most modern microscopes have oculars that are well corrected for distortion.

Chromatic aberration arises because all wavelengths of light are not brought to focus at the same distance from the optical center of a lens (longitudinal chromatic aberration). If longitudinal chromatic aberration is serious and

broad wavelength spectrum illumination is used, one can observe color fringes around the image of a subject. If a black-and-white photograph is taken of the image, it will appear slightly out of focus. Lateral chromatic aberration results from a lens producing differing magnifications for differing wavelengths of light. A simple solution is to use a narrow-band-pass filter to make the illumination nearly monochromatic; however, this precludes color photography, particularly at higher magnifications. Fortunately, lens formulas exist for the correction of chromatic aberration. Such formulas employ optical glasses of different indices of refraction.

The cost of making lenses corrected to various degrees for chromatic aberrations increases dramatically as the amount of correction increases. From the amount of correction one finds that there are several classes or names for objective lenses: nonachromatic, achromatic, semi-apochromatic, and apochromatic. For work in the materials science and engineering field, non-achromatic lenses are not used.

Achromatic lenses are corrected for spherical aberration such that all rays within a limited wavelength range of the spectrum (yellow-green: 500 to 630 nm) are brought into focus essentially at the same position on the optical path. It is common practice to make chromatic correction for two specific wavelengths within this range. Correction for spherical aberration is also made for one wavelength within the range. Achromatic lenses are useful at low to medium magnifications up to 500× magnification, provided that for photography one uses black-and-white film and also a green filter to provide illumination in the yellow-green portion of the spectrum.

Improved correction for chromatic aberration can be obtained by including fluorite (LiF) lens elements in the objective (Kreidl and Rood, 1965). LiF is used because of its higher index of refraction than the normal borosilicate crown glass elements. The effect of proper optical design using fluorite elements in conjunction with borosilicate elements is to shift the focal plane of the longer wavelengths toward that of the image formed by the shorter wavelengths. Objective lenses thus corrected for chromatic aberrations are called fluorites or semiapochromatics. Such objectives provide good correction over the spectral range of 450 to 650 nm.

Apochromatic lenses are corrected for chromatic aberrations at three specific wavelengths and simultaneously for spherical aberration at two specific wavelengths. Wavelengths in the range of 420 to 720 nm are brought to the same focus. To provide optimal correction for spherical aberration, apochromatic lenses are somewhat undercorrected for color. Thus, they should be used with compensating oculars. Apochromatic lenses are of special importance to high magnifications (>800×) and for color photography.

No discussion of objective lenses is complete without comments on working distance and on depth of focus. Normal working distances (the distance between the front element of an objective lens and the specimen) are rather small even for relatively low-magnification objectives. The working distance may range from ~1 cm for a 5× objective to a fraction of a millimeter for a 100× objective. Objective lenses designed to have larger working distances for special applications are available, but they are resolution-limited due to their smaller numerical apertures. Consequently, long-working-distance objectives are not employed for high-magnification work.

Depth of focus is a property of an objective lens related to its numerical aperture. The depth of focus is the distance normal to the specimen surface that is within acceptable focus when the microscope is precisely focused on the specimen surface. Depth of focus is of concern when the specimen surface is not perfectly flat and perpendicular to the optic axis. In practice, the preparation of specimens to obtain flatness, or the depth of etching, become important factors in the utility of a given objective lens. Based on a criterion devised by Rayleigh, the depth of focus is given by $d \approx \pm n\lambda/(\text{N.A.})^2$, where d is the depth of focus, n is the index of refraction for the medium occupying the space between the objective lens and the specimen, λ is the wavelength of illumination, and N.A. is the numerical aperture (Gifkins, 1970).

Thus, for a 0.65 N.A. objective operating in air with illumination of 530-nm wavelength, the depth of focus is ~2.35 times the wavelength of light. This means that the surface of the specimen must be flat to within ~4.7 times the wavelength of light or, in this case, 2500 nm.

Oculars (eyepieces) used in modern microscope are of four main types: Huygenian, Ramsden (orthoscopic), compensating, and wide-angle (Restivo, 1992).

Huygenian oculars (Fig. 2A) consist of two simple lenses fitted with a fixed-field diaphragm located between the field lens and the eye lens. A real image is focused at the plane of the field diaphragm. Such oculars are basic low-magnification eyepieces with little correction for objective lens aberrations.

Ramsden oculars (Fig. 2B) contain the same components; however the field lens is in an inverted position and the field diaphragm is located below the field lens. As in the Huygenian ocular, a real image is located at the plane of the field diaphragm. Ramsden oculars are normally used for intermediate magnifications.

Compensating oculars have the greatest amount of correction for chromatic variations in magnification, as well as curvature of field. These eyepieces are recommended for use at high magnification and for color photomicrography with apochromatic objectives.

Wide-angle oculars in modern instruments are also designed as high-eyepoint lenses, giving a much larger relief distance between the eye lens and the eye. They contain many lens elements and their design may vary considerably from manufacturer to manufacturer. Being highly corrected for chromatic aberration and for distortion, they are often the only oculars supplied with a microscope, particularly when the instrument is fitted with flat-field objectives.

Practical Adjustment of the Microscope

Simplified schematic diagrams of transmitted- and reflected-light microscopes are given in Figure 3. Experienced microscopists have learned that the most successful use of any microscope begins by following the light path

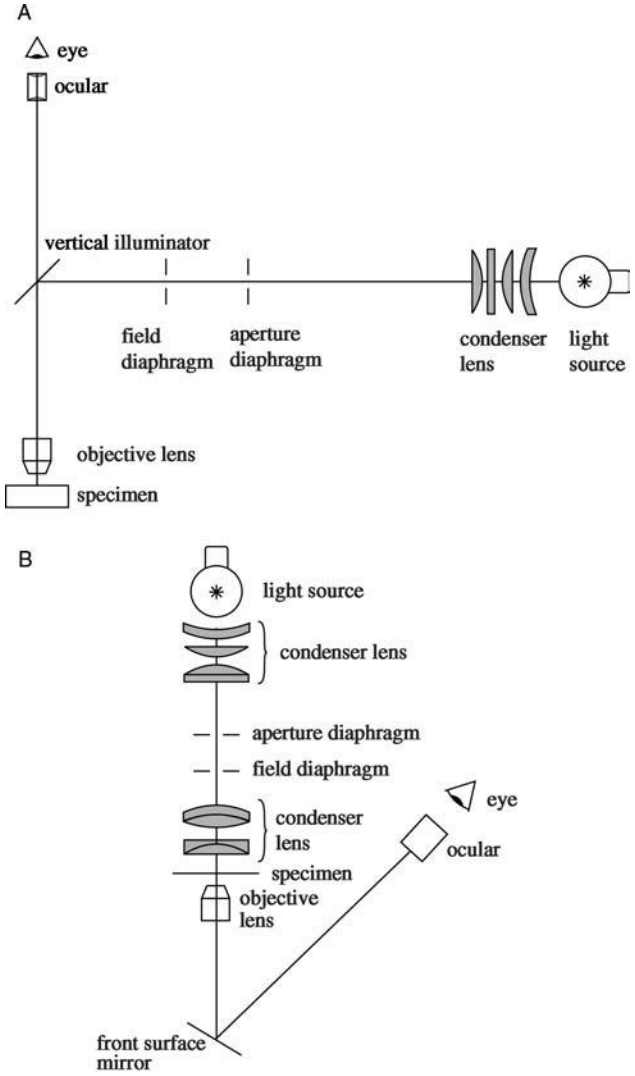


Figure 3. Schematic diagrams of (**A**) inverted reflected-light and (**B**) transmitted-light microscopes. The important components are labeled. Note that the transmitted-light microscope (panel B) does not have a vertical illuminator.

through the instrument and making necessary adjustments, starting with the source of illumination. By consistently following the sequence, the chance of failure in the form of a poor photomicrograph or poor visual image is greatly reduced.

Illumination of the specimen is accomplished by two or more components including the lamp and a condenser or collector lens. Sometimes, the aperture diaphragm, field diaphragm, heat-absorbing filter, narrow band-pass filter, and neutral-density filter are also included as components of the illumination system. This author chooses not to include the additional components as parts of the illumination system, although some microscope makers mount them in a single modular unit of the instrument.

Proper centering and focusing of the light from the illuminator is essential to obtaining satisfactory results with any optical microscope. The image of the lamp filament or arc must be focused at the plane of the aperture diaphragm. In some instruments, the light path is sufficiently open so that one can visually inspect the focus of the lamp on a closed-aperture diaphragm. In closed systems, specific instructions for obtaining proper focus are usually provided with the instrument. Centering of the light source is accomplished by adjustments that laterally translate the lamp in two directions. A focusing adjustment moves the lamp closer to or further away from the condenser to focus the lamp at the plane of the aperture diaphragm.

The aperture diaphragm must be properly adjusted to obtain the optimum resolution commensurate with image contrast, for either a transmission- or reflected-light microscope. The full numerical aperture of the objective lens must be utilized to obtain optimum resolution. From a practical standpoint, the aperture diaphragm is interposed in the optical path to control the diameter of the pencil of light so that it just fills the full circular area of the rear lens group of the objective lens. In a transmitted-light microscope the aperture diaphragm is located in the condenser lens barrel, while in a reflected-light microscope the aperture diaphragm is located in the optical path just beyond the condenser (collector) lens of the light source. In both instruments, the image of the diaphragm is focused in a plane coincident with the rear element of the objective lens. The aperture diaphragm is always partially closed to reduce internal light scattering within the microscope, a circumstance that degrades contrast in the image. In practice, the method for obtaining the optimum setting of the aperture diaphragm is to open the diaphragm fully, bring the specimen into focus, and then close the diaphragm slowly until the image begins to dim perceptibly. At this point the aperture is opened slightly to bring the image back to full brightness.

The image of the field diaphragm is focused on a plane coincident with the plane of the specimen. Its function is to reduce extraneous light scatter between the specimen surface and the front element of the objective in order to enhance contrast. To adjust the field diaphragm, one starts with it in the full-open position and gradually closes it while viewing the focused specimen either through the eyepieces or on a photographic viewing screen. When the image of the diaphragm is observed to just intrude into the field of view or into the photographic field, it should be opened slightly.

Vertical illuminators are required for reflected-light microscopes, because the specimens are opaque and the objective lens must act both to supply light to, and to collect reflected light from, the specimen. The light path must be passed to the specimen through the objective lens, then allow the reflected image-forming light to pass back through the objective to the ocular. From Figure 3A, note that the light path is diverted 90° from the horizontal to the specimen, which rests inverted on the stage (inverted microscope). This arrangement is of particular utility in using specimens that must have prepared surfaces, because it does not require that the specimen have a back surface exactly parallel to the prepared surface. Several schemes have been used for vertical illuminators, including various prism configurations and thin, optically flat glass plates. Of these, the thin, optically flat glass plate

(or disc) is almost universally used in recent instruments. Generally, vertical illuminators have no adjustment controls except for a provision for dark field illumination, a topic that is discussed in REFLECTED-LIGHT OPTICAL MICROSCOPY.

The objective lens requires only two adjustments: centering the optic axis of the lens with the optic axis of the instrument and focusing the lens by moving it closer to or further away from the specimen. Focusing the objective is as simple as moving the focusing adjustments while viewing the specimen through the eyepieces and observing when the specimen appears to be sharp. A number of instruments have their objective lens mounted in a turret to facilitate changing from one objective to another. Many instruments are said to have "parafocaled" objectives. If the objectives are parafocaled, one may readily switch objectives without having to refocus the instrument. In practice, however, it is wise to use the fine-focus adjustment each time an objective is changed in order to assure that the optimum focus is obtained.

Centering of the objective lens on an instrument with an objective turret is accomplished by centering on the objective through the centering of the axis of rotation of the turret. This involves moving the axis of rotation of the turret by shifting it in one or two orthogonal directions of a plane normal to the axis of rotation. Such an operation is not recommended except when performed by competent service personnel. Fortunately, once a turret is centered and locked into the instrument, frequent centering is not required. Other instruments, in which objectives are placed in the instrument individually, require that each lens be centered individually by adjustment set screws on the mounting plate of each lens. Often the instrument contains one "standard" objective that is centered at the factory in the final fitting stages of instrument assembly. The mounting plate of this objective has no centering adjustments. Other objectives may be centered individually using the "standard" lens as a reference, provided the instrument is fitted with a rotating stage that is also centered with the optic axis of the instrument. A simple method involves starting with the "standard" objective, aligning a reflective cross-hair (or reflective point target) on the stage with the center of the field as viewed through the ocular, and making certain that the stage is itself centered. The stage is then rotated to ascertain that the cross-hair stays in the center of the field as the stage is rotated. Without moving the stage or the cross-hair, a centerable objective is substituted for the "standard" objective. If the objective is centered, the cross-hair will again fall in the center of the field and will remain there when the stage is rotated. To center the objective, it will be necessary to use the adjustment set screws on the mounting plate to move the optic axis of the objective in x and y directions in a plane normal to the optic axis. A number of instruments include special accessories for centering the objectives.

Ocular adjustment amounts only to focusing the ocular. Instruments with a single eyepiece generally do not have an adjustment for focus, because the eyepiece is set at its proper focal distance from the objective. Since most microscopes now have binocular eyepieces, some care must be taken to assure that the eyepieces are correctly focused to match the eyes of the user. One eyepiece has a focusing ring, while the other is fixed. Depending on the instrument, focusing methods may vary; however, they are all variations on the method outlined in the following. A stage micrometer, or other similar device, is placed on the stage and brought into focus with the fixed eyepiece using the objective focusing adjustment. Once this is done, the eyepiece focusing ring should be used to focus the image for the other eye. This method accounts for differences between the two eyes, and is necessary to reduce eye strain from long sessions with the microscope.

It is important to emphasize that if the microscopist normally wears eyeglasses, they should be worn while using a microscope. In recent years wide-field objective lenses and matching wide-field, high-eyepoint oculars have been introduced. Matching optics of this type provide a wider field of view of the specimen and are also particularly suited to users who wear glasses. Multifocal eyeglasses (bifocal or trifocal) present special problems that can be overcome by the adaptability of the microscopist.

Autofocusing has yet to be applied to microscopes except in the case of instruments designed to make precise measurements. Such devices are only of use when one of several turret-mounted objective lenses is brought into sharp focus. The autofocusing feature will then cause each of the other lenses in the turret to be brought into sharp focus when they are rotated into the optic axis of the microscope. Critical focusing depends on the skill of the microscopist in being able to judge when sharp focus is attained. Focusing for edge sharpness could probably be developed for instruments used in conjunction with image analysis computers, wherein the focusing would be accomplished by feedback of the contrast gradient at a sharp edge. However, obtaining the best overall focus for the entire field of view might be difficult, either because of a limited range of the gray scale or because of the lack of flatness of specimens.

Devices for calibrating resolution or the depth of focus are generally not used. Observable degradation of the image is a signal to inspect the optical components for contaminants and for physical defects such as scratched or etched surfaces. Regular annual service by a competent service technician is recommended, to maintain the microscope in optimum working condition.

Damage from contamination and damage to objectives is, for the most part, up to the microscopist in maintaining a clean environment for the microscope. The use of appropriate lubricants for the mechanical parts of the instrument is of paramount importance. Physical damage of the front element of the objective lens comes primarily from two sources: (1) contact of the front element with the specimen and (2) etching of the front element by chemical reagents used in specimen preparation. Careful operation by the microscopist is the most effective way to keep the objective lens from contacting the specimen; however, some microscopes, usually upright instruments, employ a stage lock to prevent the objective from contacting the specimen. To prevent contamination of the objective lenses with specimen-preparation reagents, it is essential that specimens be thoroughly cleaned and dried

precipitates. When some portion of the image appears to be in focus, while others appear to be poorly focused, the problem is usually associated with lack of flatness of the specimen surface. In addition, one should pay attention to the action of polishing abrasives on the specimen surface. Some combinations of abrasive materials result in undesired chemical attack of the specimen surface, and the true structure of the specimen may not be revealed. Poor etching (over- or underetching) is a major concern to the metallographer. Overetching often causes what appears to be lack of resolution; fine features can be difficult or impossible to resolve. Underetching may leave some fine details not revealed at all. A better understanding of some aspects of metallographic specimen preparation is given in SAMPLE PREPARATION FOR METALLOGRAPHY.

A practical guide to the setting of the controls is given in OPTICAL MICROSCOPY. Some of the common errors in making settings that degrade resolution of the image are: (1) too small an aperture diaphragm setting, (2) incorrect selection of objective lens, and (3) too high a magnification. Closing the aperture diaphragm beyond the point where the rear element of the objective lens is just covered with light will not allow the objective lens to collect all the diffraction information needed to realize the full resolution capabilities of the lens. The numerical aperture (N.A.) of the objective lens must be consistent with the intended magnification of the final image. As a practical upper limit, the final magnification should be no greater than 1000 times the numerical aperture of the objective lens. For a 0.65 N.A. objective lens, the final magnification should be 650× or less.

Dirty or damaged optical components seriously degrade image quality. Dirt and dust on lens, mirror, or prism surfaces, as well as degradation of lens coatings, are of primary concern. It is wise not to attempt optical surface cleaning other than that which can be done with a soft brush or light blast of air. It is well worth the expense to have a skilled service person thoroughly clean the microscope on an annual basis.

The forgoing discussions are not intended to be a comprehensive treatment of reflected-light microscopy. A number of interesting special techniques have been excluded because they require specialized instrumentation. Almost all of the illumination modes and image-enhancement techniques discussed can be done with current research-grade metallographs. One feature that most modern instruments do not include is a precision rotating stage that allows for rotation of the specimen without disturbing the focus of the instrument. Such a stage was incorporated on instruments manufactured through the mid-1970s. Stage rotation, wherein the specimen remains normal to the optic axis of the instrument, greatly enhances the ability to use polarized light, sensitive tint, and differential interference contrast.

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DeHoff and Rhines, 1968. See above.

A textbook of the well-established fundamentals underlying the field of quantitative microscopy, which contains the details of applied measurement techniques.

Gifkins, 1970. See above.

Chapter 6 of this book provides an excellent treatment of interference techniques.

Kehl, G.L. 1949. The Principles of Metallographic Laboratory Practice. McGraw-Hill, New York.

Excellent reference on microscopes and photomicrography for use by technicians, although some of the instrumentation is dated.

Vander Voort, 1984. See above.

Chapter 4 of this book is a particularly good overview reference for reflected-light microscopy.

INTERNET RESOURCES

http://microscopy.fsu.edu/primer/resources/tutorials.html

List of educational resources for microscopy.

http://www.people.virginia.edu/jaw/mse310L/w4/mse4-1.html

Optical micrography reference.

RICHARD G. CONNELL, JR.
University of Florida
Gainesville, Florida

PHOTOLUMINESCENCE SPECTROSCOPY

INTRODUCTION

Photoluminescence (PL) spectroscopy is a powerful technique for investigating the electronic structure, both intrinsic and extrinsic, of semiconducting and semi-insulating materials. When collected at liquid helium temperatures, a PL spectrum gives an excellent picture of overall crystal quality and purity. It can also be helpful in determining impurity concentrations, identifying defect complexes, and measuring the band gap of semiconductors. When

performed at room temperature with a tightly focused laser beam, PL mapping can be used to measure micrometer-scale variations in crystal quality or, in the case of alloys and superlattices, chemical composition.

The information obtainable from PL overlaps with that obtained by absorption spectroscopy, the latter technique being somewhat more difficult to perform for several reasons. First, absorption spectroscopy requires a broadband excitation source, such as a tungsten-halogen lamp, while PL can be performed with any above-band-gap laser source. Second, for thin-film samples absorption spectroscopy requires that the substrate be etched away to permit transmission of light through the sample. Third, for bulk samples the transmitted light may be quite weak, creating signal-to-noise ratio difficulties. On the other hand, absorption spectroscopy has the advantage of probing the entire sample volume, while PL is limited to the penetration depth of the above-band-gap excitation source, typically on the order of a micrometer.

The areas of application of PL also overlap somewhat with Raman spectroscopy (RAMAN SPECTROSCOPY OF SOLIDS), but the latter is much harder to perform because of the inherent weakness of nonlinear scattering processes and because of its sensitivity to crystallographic orientation. On the other hand, this sensitivity to orientation gives Raman spectroscopy a capability to detect variations in crystallinity that PL lacks.

There are also more elaborate techniques, such as photoluminescence excitation (PLE), time-resolved PL, and resonant Raman scattering, but these techniques are more useful for basic scientific investigations than for routine crystal characterization.

PRINCIPLES OF THE METHOD

When a semiconductor is illuminated by a light source having photon energy greater than the band-gap energy (E_g) of the material, electrons are promoted to the conduction band, leaving holes behind in the valence band. When an electron-hole pair recombines, it emits a photon that has a wavelength characteristic of the material and the particular radiative process. Photoluminescence spectroscopy is a technique for extracting information about the electronic structure of the material from the spectrum of light emitted.

In the basic process of PL at low temperature (liquid nitrogen temperature and below), an incoming photon is absorbed to create a free electron-hole pair. This transition must be essentially vertical to conserve crystal momentum, since the photon momentum is quite small. At cryogenic temperatures the electrons (holes) rapidly thermalize to the bottom (top) of the conduction (valence) band by emitting phonons. For this reason there is typically no observable luminescence above the band-gap energy. Once they have relaxed to the band edges, an electron and hole can bind together to form an exciton, which then recombines radiatively as described below. Alternatively, the electron or hole (or both) can drop into a defect state by nonradiative recombination and then undergo radiative recombination.

Excitons and Exciton-Polaritons

It is impossible to analyze luminescence spectra without considering the role of excitons. Formally an exciton must be thought of as an additional bound state of the crystal that appears when the Coulomb interaction between electrons and holes is incorporated into the crystal Hamiltonian, but there is also a simple, mechanistic picture that leads to the same result. The exciton can be thought of as a hydrogenlike atom that results when the electron and hole bind together. The binding energy of an exciton is then given by

$$E_n = -\frac{E_0}{n^2} \qquad E_0 > 0 \qquad n = 1, 2, 3 \tag{1}$$

where E_0 is the free exciton binding energy. The total energy of the exciton is $E_g + E_n < E_g$, so when the exciton recombines, the energy of the emitted photon is below E_g.

This simple, hydrogenlike picture of the exciton is actually an oversimplification. In reality, the free exciton should be thought of as a polarization wave that can propagate through the crystal. This polarization wave can interact with an electromagnetic wave to produce *exciton-polariton* states. The dispersion curve for these states has two branches, as illustrated in many texts in solid-state and semiconductor physics (e.g., Yu and Cardona, 1996, p. 273). Given this continuum of states, the polariton emission spectrum can be quite complex, the peak positions being determined by the relative lifetimes of states at various positions on the dispersion curve. In particular, polaritons near the transverse free exciton energy have particularly long lifetimes, creating a "bottleneck" that leads to accumulation of exciton population. Hence there is typically a peak at the free exciton energy, which can be thought of as a "pure" free exciton peak and is often labeled as such in the literature and in this unit.

Bound Excitons

At low temperatures excitons can bind to donor or acceptor sites via van der Waals forces to form states of lower energy than the polariton states. Since bound exciton states are localized, they have no dispersion. The dissociation energy for a bound exciton is dependent on the species of donor or acceptor; this fact is sometimes helpful in identifying impurities. Figure 1 shows donor- and acceptor-bound exciton luminescence for CdTe.

Alloy Broadening

In an alloy, random microscopic composition variations in the crystal cause a broadening in the energy spectrum of electronic states. Figure 2 shows the exciton spectrum for $Cd_{0.96}Zn_{0.04}Te$. Much of the fine structure that was apparent for pure CdTe has been obscured by alloy broadening.

Phonon Replicas

There may also be phonon replicas of any of the aforementioned peaks. That is, the system may emit part of its energy as one or more phonons—typically longitudinal-

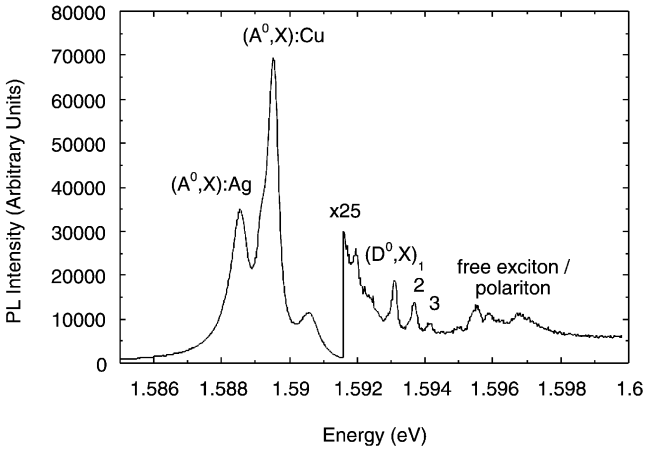


Figure 1. Free and bound exciton luminescence of CdTe at 4.2 K showing bound exciton luminescence from two acceptor species.

optical (LO) phonons—and the remainder as a photon. The emitted photon is thus of longer wavelength than that for the no-phonon process. In some cases the emitted phonon corresponds to a local vibrational mode of the defect to which the exciton is bound; this provides another means for identifying defects. In Figure 2, the first phonon replica of the free exciton luminescence is visible. The phonon replicas are less well-resolved than the no-phonon lines, because of the slight dispersion in the LO phonon energy. The LO phonon energy can be read from the spacing of the peaks as ~22 meV.

Donor-Acceptor and Free-to-Bound Transitions

Some electrons and holes drop by nonradiative processes from the conduction and valence bands into donor and acceptor states, respectively. When these charge carriers recombine, they emit photons of lower energy than do excitonic or direct band-to-band transitions. Possible transitions include those in which a conduction band electron

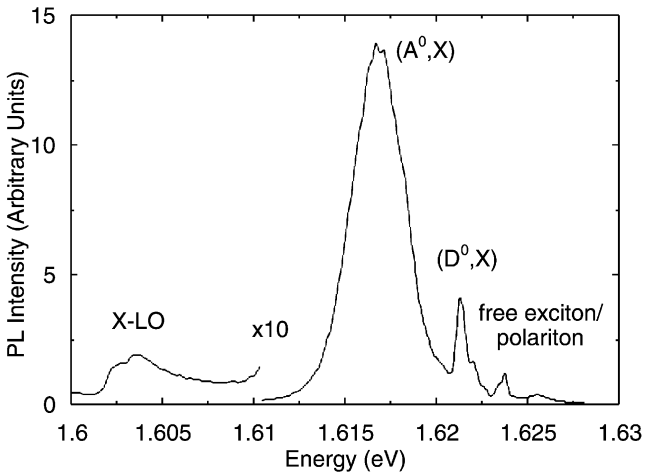


Figure 2. Exciton luminescence of $Cd_{0.96}Zn_{0.04}Te$ at 4.2 K. Many of the peaks that are present in the CdTe spectrum are not resolvable because of alloy broadening.

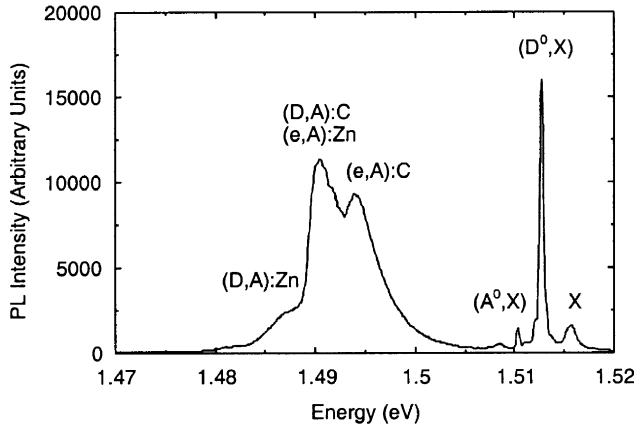


Figure 3. PL spectrum of semi-insulating GaAs at 4.2 K showing separate donor-acceptor and band-to-acceptor peaks for C and Zn.

recombines with a hole in an acceptor level (e,A), an electron in a donor level combines with a hole in the valence band (D,h), and an electron in a donor level recombines with a hole in an acceptor level (D,A). The (D,A) luminescence is below the band gap by the sum of the donor and acceptor binding energies minus the Coulomb energy of attraction between the ionized donor and acceptor in the final state:

$$h\nu = E_g - (E_A + E_D) + \frac{e^2}{\varepsilon r_n} \quad \text{(Gaussian units)} \qquad (2)$$

where r_n is the distance between the nth nearest neighbor donor and acceptor sites and can take on only those values consistent with the crystal lattice. The discrete values of r imply that there are a large number of separate (D,A) peaks, but for relatively large values of n, they merge into a single broad band. With high-resolution measurements on high-quality crystals at low temperature, it is sometimes possible to detect separate peaks for low values of n (Yu and Cardona, 1996, p. 346).

There can also be multiple donor or acceptor levels in a given material, each exhibiting its own set of peaks in the spectrum. A well-known example is C and Zn in GaAs, which give rise to two sets of overlapping (e,A) and (D,A) peaks (Stillman et al., 1991). A representative 4.2 -K GaAs spectrum is shown in Figure 3.

Transitions Involving Defect Levels

Many kinds of defects produce energy states closer to the middle of the band gap than the shallow donor and acceptor states mentioned above. Recombination processes involving these defect levels result in emission of longer-wavelength photons. Because these defect states are localized, they have broad energy spectra. Also, they tend to be strongly coupled to the lattice and hence have a large number of phonon replicas that often smear together into a single broad band. An example CdZnTe spectrum is shown in Figure 4. The band at ~1.4 eV is believed to be related to the A-center (Cd vacancy plus donor), and the band at 1.1 eV is believed to be related to Te vacancies.

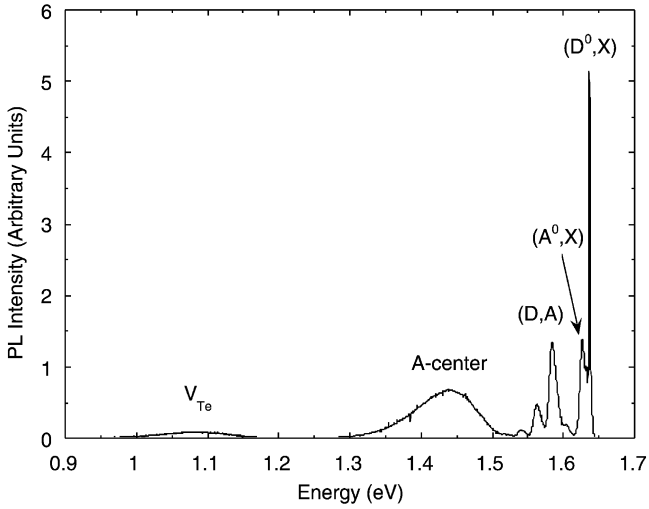


Figure 4. PL spectrum of $Cd_{0.9}Zn_{0.1}Te$ at 4.2 K showing excitonic, donor-acceptor, and deep-level luminescence.

Band-to-Band Recombination

At room temperature there is a nonzero occupancy of electron states near the bottom of the conduction band and hole states near the top of the valence band. Hence, electron-hole pairs can recombine to emit photons over a range of energies, producing a broad band rather than a sharp peak. Donor, acceptor, and defect states are generally fully ionized at room temperature and do not contribute significantly to the observed spectrum.

Band-to-band recombination produces a luminescence spectrum that is the product of the joint density of states and the Fermi-Dirac distribution (see Bhattacharya, 1994):

$$I(E) \propto \frac{\sqrt{E - E_g}}{1 + \exp[(E - E_g)/kT]} \tag{3}$$

Observed room temperature luminescence typically does not follow Equation 3 except at the upper end of the energy range. Equation 3 has a sharp lower cutoff at the band-gap energy, while the measured spectrum has a long tail below the peak. This is an indication that the spectrum is only partly due to band-to-band recombination and also includes excitonic and phonon-assisted transitions (Lee et al., 1994b).

PRACTICAL ASPECTS OF THE METHOD

Experimental Setup

The apparatus for low-temperature PL can be set up easily on an optical table. The sample is enclosed in an optical cryostat and illuminated with an above-band-gap light source, in most cases a laser. An optical band-pass filter is typically used to remove unwanted plasma lines from the laser emission, and a low-pass filter is often placed at the entrance of the monochromator to remove stray light at shorter wavelengths than the range of interest. Both of these sources of extraneous light can show up in the spectrum in second order. The light emitted by the sample is collected with a collimating lens and then focused onto the entrance slits of the monochromator with a second lens.

A photon detector, typically either a photomultiplier tube (PMT) or a positive-intrinsic-negative (PIN) diode detector, is placed at the exit of the monochromator. The output signal of the detector is routed either to a photon counter or to a lock-in amplifier. In the latter case, a light chopper is used to synchronize the signal with a reference. In the photon-counting mode, it is desirable to cool the detector with liquid nitrogen to reduce noise; N_2 gas must then be allowed to flow over the face of the detector to keep it from frosting.

The instrumentation for room temperature PL mapping is similar, with the addition of an *x*-*y* translation stage. Generally, the optics are kept fixed and the sample translated. The excitation source is focused onto the sample with a microscope objective, which also serves to collimate the luminescence. A beam splitter or a mirror containing a small hole is used to allow transmission of the laser beam while reflecting the luminescence into the monochromator.

An alternative scheme for PL uses an optical multichannel analyzer. In this case the monochromator disperses the light onto an array of charge-coupled devices (CCDs) so that the entire spectrum is captured at once. This configuration has the advantage of greater speed, which is especially valuable in room temperature mapping, but it generally suffers from reduced wavelength resolution.

Following is a list of instrumentation and representative manufacturers for low-temperature PL along with some of the requirements and considerations for each apparatus:

1. *Detectors* (Hamamatsu, Phillips, Burr-Brown, Electron Tubes, Oriel, EG&G, Newport)—The principal trade-off is between bandwidth and sensitivity. The need for cooling is a further consideration. Photomultiplier tubes are unmatched in sensitivity but typically have limited long-wavelength response. Most PMTs can be operated at room temperature in the current mode but may require cooling for optimal performance in the photon-counting mode. Narrow-band-gap semiconductor detectors, such as germanium PIN diodes, have better long-wavelength response but poorer sensitivity at shorter wavelengths. They generally must be cooled with liquid nitrogen.
2. *Spectrometers/monochromators* (Jobin Yvon-Spex, Oriel, Jarrell Ash, Acton Research, McPherson, Ocean Optics, CVI Laser)—For high resolution, a large, double-grating spectrometer is required. For lower resolution applications, compact, inexpensive prism spectrometers with integrated CCD detectors are available.
3. *Lasers* (Coherent, Uniphase, Kimmon, Spectra-Physics, Newport, Melles Griot, Anderson Lasers)—The principal requirement is that the photon energy be higher than the band gap of the material

at the temperature at which the measurements are made. For direct band-gap materials, output power of a few milliwatts is typically sufficient. For indirect band-gap materials, higher power may be desirable. Single-line operation is preferred, since it permits the use of a laser line filter to remove extraneous lines that can appear in the measured spectrum in second order.

4. *Lock-in amplifiers/photon counters* (EG&G, Stanford Research, Oriel)—Both analog and digital lock-in amplifiers are available, the digital instruments generally having superior signal-to-noise ratios.
5. *Complete PL spectroscopy/mapping/imaging systems* (Phillips Analytical, Perkin Elmer, Oriel, ChemIcon)—Some systems are configurable with various laser modules, gratings, and detectors to be optimized for the application.

Use of Low-Temperature PL for Crystal Characterization

The most basic use of low-temperature PL spectra is as an indicator of overall crystal quality. The principal indicator of quality is the ratio of excitonic luminescence to donor-acceptor and deep-level luminescence. Figure 5 shows PL spectra for two CdZnTe samples of different quality. In the high-quality sample, the spectrum is dominated by a donor-bound exciton peak (D^0,X), while in the poor-quality sample the excitonic luminescence is outweighed by donor-acceptor and defect bands.

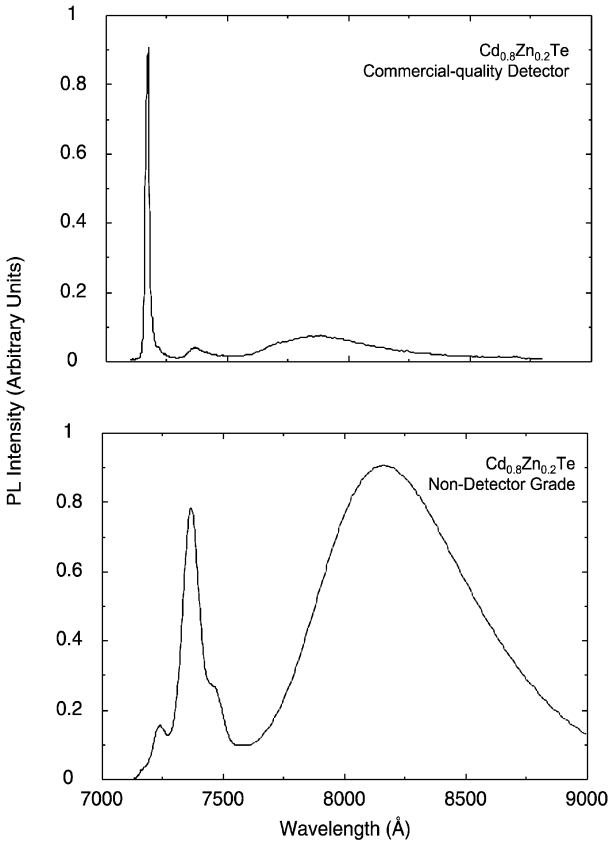


Figure 5. The 4.2-K PL spectra for $Cd_{0.8}Zn_{0.2}Te$ samples of differing quality. In the high-quality (commercial-quality detector grade) sample (top), the spectrum is dominated by excitonic luminescence. In the poor-quality (non-detector grade) sample the donor-acceptor and deep-level luminescence dominate.

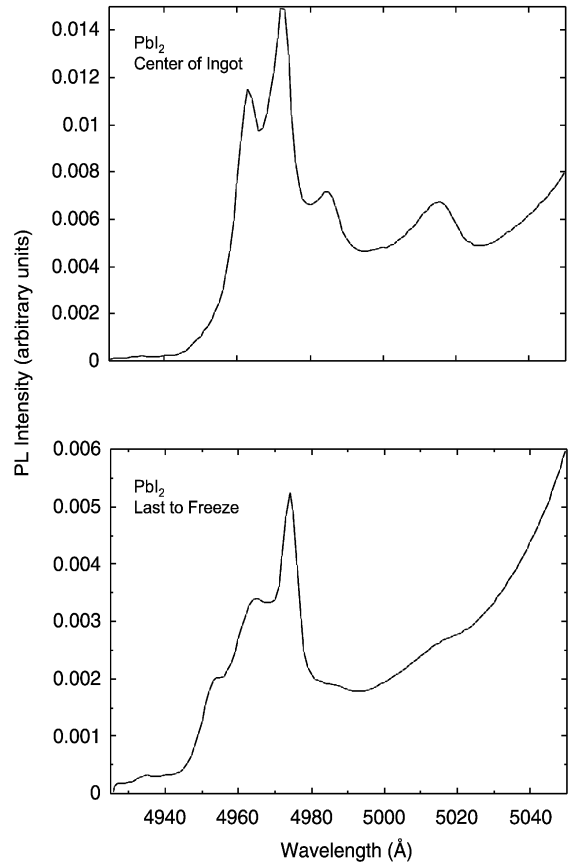


Figure 6. 4.2 K PL spectra for PbI_2 samples from center (top) and last-to-freeze portions of ingot. The spectrum for the sample from the middle of the ingot is sharper, indicating better crystal quality.

A second measure of crystal quality is the sharpness of the spectrum—that is, the extent to which adjacent lines can be resolved. Figure 6 shows near-band-edge PL spectra for two PbI_2 samples, one from the center of the ingot and the other from the last-to-freeze portion. The spectrum of the sample from the center is sharper, indicating a more uniform crystal potential on a microscopic scale.

Aside from indicating overall crystallinity, the low-temperature PL spectrum can sometimes be helpful in identifying specific impurities. The example of the C and Zn (D,A) bands in GaAs was given in Figure 4, but the bound exciton luminescence can also be used. For example, in the CdTe spectrum shown in Figure 1, the (A^0,X) peak at 1.5895 eV is due to Cu, while that at 1.5885 eV is due to Ag (Molva et al., 1984).

The ratio of bound to free exciton luminescence can also be used to measure impurity concentrations, but this application requires careful control over excitation intensity and sample temperature (Lightowlers, 1990). It also requires that calibration curves be generated based on chemical analysis.

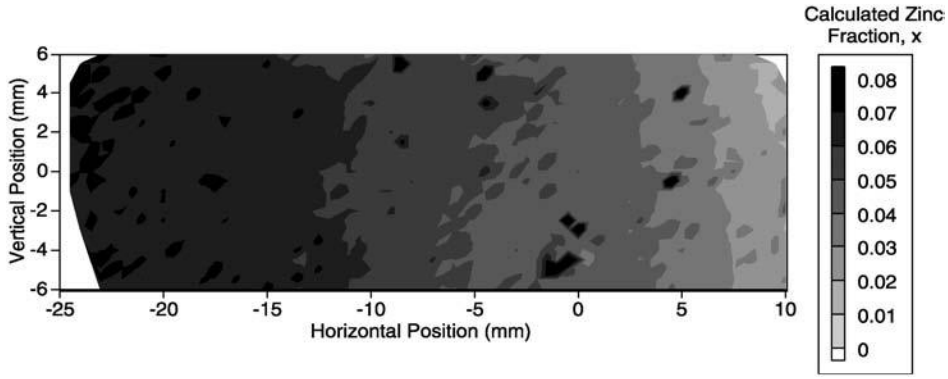


Figure 7. Room temperature PL composition map of $Cd_{1-x}Zn_xTe$ showing segregation of zinc along the growth axis.

Uses of Room Temperature PL Mapping

One application of room temperature PL is composition mapping of alloys. Figure 7 shows a PL composition map for CdZnTe grown by a high-pressure Bridgman method. Segregation of zinc along the growth axis is clearly visible.

In addition to the systematic variation in composition along the growth axis, local variations are sometimes seen. Figure 8 shows an example in which minute striations are present, perhaps due to temperature fluctuations during growth. Note that the dark bands are only ~20 μm wide, demonstrating that high spatial resolution is important.

Photoluminescence mapping is also useful for studying structural defects. PL maps of CnZnTe often contain isolated dark spots that can be attributed to inclusions, possibly of tellurium or zinc. High-spatial-resolution PL maps of these inclusions show a region of near-zero PL intensity surrounded by a region in which the band gap is shifted to higher energy—hence the dark spots in the larger-scale maps. It is unclear whether the shift is caused by gettering of zinc around the inclusion or by strain.

METHOD AUTOMATION

Complete automated systems for PL spectroscopy and mapping are now commercially available. If a system is constructed from scratch, it is a relatively simple matter to integrate GPIB-controllable equipment using general-purpose laboratory software packages.

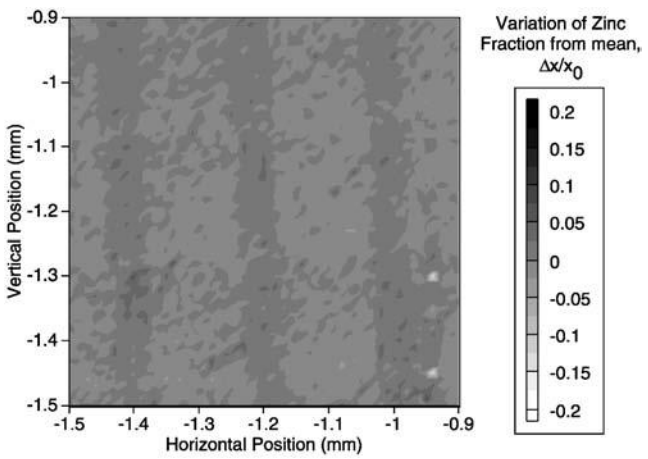


Figure 8. Room temperature PL composition map of $Cd_{1-x}Zn_xTe$ showing growth striations.

DATA ANALYSIS AND INITIAL INTERPRETATION

Identification of Bands in a Low-Temperature PL Spectrum

Qualitative interpretation of a low-temperature PL spectrum is quite simple if there are reliable sources in the literature to identify the origins of the various bands. If there are bands of uncertain origin, the intensity and temperature dependence of the spectrum can give clues to their nature.

As the sample temperature is raised, the luminescence is quenched as the centers involved in the transition are ionized by thermal excitation. The activation energy or energies of the level can then be determined by a curve fit to the following equation, which follows from Boltzmann statistics (Bimberg et al., 1971):

$$I(T) = I(0)\left[1 + C_1\exp\left(-\frac{-E_1}{kT}\right) + \cdots + C_n\exp\left(-\frac{-E_n}{kT}\right)\right]^{-1} \quad (4)$$

where $I(T)$ is the integrated intensity of the band at absolute temperature T, k is Boltzmann's constant, E_i is the activation energy of the ith process involved in the dissociation, and C_i is formally the degeneracy of the level but in practice is left as a fitting parameter. Most processes can be modeled as one- or two-step dissociations. An example of a one-step dissociation is that of a free exciton into a free electron and hole, whereas the prototypical two-step dissociation is bound exciton quenching, in which the exciton first dissociates from the donor or acceptor to form a free exciton, which then dissociates in a second step. This difference in temperature dependence is one means of distinguishing between free and bound excitons in an unfamiliar spectrum.

Temperature dependence can also be used to distinguish between (D,A) and (e,A) transitions, since the former, which involve a shallow donor, will be quenched quickly as the donors are ionized.

The dependence of the spectrum on the excitation intensity can be used to distinguish between excitonic and

donor-acceptor bands. In both cases the luminescence intensity varies with the excitation intensity as a power law, but for excitonic lines the exponent is >1.0, whereas for donor-acceptor and free-to-bound transitions it is <1.0 (Schmidt et al., 1992).

One additional technique that can help to identify peaks is infrared quenching. As the sample is subjected to below-band-gap infrared radiation in addition to laser excitation, donors and acceptors are ionized, causing quenching of (D^0,X) and (A^0,X) transitions and an increase in (D^+,X) luminescence. It has been shown that (A^-,X) complexes cannot exist in most semiconductors (Hopfield, 1964), so that this provides another way of distinguishing between donor- and acceptor-bound excitons. Infrared quenching has also been used quantitatively to measure impurity concentrations (Lee et al., 1994a).

Curve Fitting in Room Temperature PL Mapping

Extracting material parameters from a room temperature PL spectrum in a systematic way requires that the calculated line shapes for various types of transitions (Bebb and Williams, 1972) be combined. This approach has been used with some success in cadmium telluride (Lee et al., 1994b), and in cadmium zinc telluride (Brunnett et al., 1999). In the alloy, variation in the band gap and exciton and phonon energies introduces additional uncertainty in the extracted composition. The composition determined from this procedure cannot be regarded as highly accurate, but variations can be detected with precision.

SAMPLE PREPARATION

Photoluminescence generally requires no special sample preparation, other than to provide a reasonably smooth surface so that a large fraction of the light is not scattered away. Thin-film samples generally can be analyzed as grown, whereas bulk samples that have been cut typically require etching to remove the damaged layer produced by the cutting.

SPECIMEN MODIFICATION

Low-temperature PL requires relatively low optical power density, so that the risk of damaging the material is minimal. Room temperature mapping, however, does use rather high power densities that cause significant sample heating. Most group II–VI and III–V compounds produce adequate luminescence at low enough intensity to avoid damaging the sample, but with less robust materials, such as PbI_2 and HgI_2, room temperature mapping may not be practical.

POTENTIAL PROBLEMS

A difficulty with extracting accurate composition information from room temperature PL spectra is that the peak position depends on the excitation wavelength and intensity. The spectrum shifts to lower energy as the intensity is raised, and the peak position is at lower energy for the shorter-wavelength light. The explanation for this effect probably lies in local heating of the sample by the rather high intensities used. The heating is greater for higher-energy (blue) photons because a greater fraction of each photon's energy is converted to phonons.

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